

Tracker SDK — Provider Abstraction Architecture

Status: descriptive architecture reference. Every box, arrow, and symbol below corresponds to a real file in the working tree (cited inline). Anything not yet implemented is marked **NOT IMPLEMENTED**. Per HelixConstitution §11.4.6 (no-guessing vocabulary), nothing here is inferred — it is read from source.

Companion docs: [docs/ARCHITECTURE.md](#) (overview), [docs/sdk-developer-guide.md](#) (the 7-step recipe

- the magnet-cache + download-validator patterns), [docs/architecture/local-stack-topology.md](#) (server-side deployment).

1. The abstraction in one diagram

The Tracker SDK is a capability-gated plugin system. A feature ViewModel talks only to LavaTrackerSdk; the SDK routes the call through the active tracker's `TrackerClient.getFeature<T>()`, which returns a feature-interface implementation **only when the descriptor declares the matching capability** (constitutional clause 6.E, "Capability Honesty").

```
graph TD
    VM["feature:* ViewModels<br/>(search_result, topic, login,<br/>>provider_config, menu ...)"]
    SDK["LavaTrackerSdk @Singleton<br/>core/tracker/client/.../LavaTrackerSdk.kt"]
    REG["TrackerRegistry / DefaultTrackerRegistry<br/>core/tracker/registry/.../DefaultTrackerRegistry.kt"]
    FAC["TrackerClientFactory (per provider)<br/>core/tracker/registry/.../TrackerClientFactory.kt"]
    CLIENT["TrackerClient (active)<br/>core/tracker/api/.../TrackerClient.kt"]
    DESC["TrackerDescriptor<br/>core/tracker/api/.../TrackerDescriptor.kt"]
    CAP["TrackerCapability enum<br/>core/tracker/api/.../TrackerCapability.kt"]
    GATE["getFeature<T>(KClass<T>)<br/>capability ∈ descriptor.capabilities?"]
    FEAT["TrackerFeature impls<br/>core/tracker/api/.../feature/*.kt"]

    VM -->|"inject @Singleton"| SDK
```

```

SDK -->|"registry.get(id, config)"| REG
REG -->|"create(config)"| FAC
FAC -->|"returns"| CLIENT
CLIENT -->|"val descriptor"| DESC
DESC -->|"capabilities: Set"| CAP
SDK -->|"client.getFeature(SearchableTracker::class) ..."| GATE
GATE -->|"declared → impl"| FEAT
GATE -->|"not declared → null<br/>(Failure outcome, never
throws)"| SDK

```

Grounding:

- LavaTrackerSdk — core/tracker/client/src/main/kotlin/lava/tracker/client/LavaTrackerSdk.kt
- TrackerClient.getFeature — core/tracker/api/src/main/kotlin/lava/tracker/api/TrackerClient.kt
- TrackerDescriptor — core/tracker/api/src/main/kotlin/lava/tracker/api/TrackerDescriptor.kt
- TrackerCapability — core/tracker/api/src/main/kotlin/lava/tracker/api/TrackerCapability.kt
- TrackerClientFactory / DefaultTrackerRegistry — core/tracker/registry/src/main/kotlin/lava/tracker/registry/

2. The contract surface (core:tracker:api)

2.1 TrackerDescriptor

A descriptor is the provider's identity + honesty contract. The fields below are verbatim from TrackerDescriptor.kt:

Field	Type	Meaning
trackerId (= id)	String	Stable id, e.g. "rutracker", "rutor", "kinozal", "nnmclub". Extends lava.sdk.api.HasId.
displayName	String	Human-readable name shown in UI.
baseUrls	List<MirrorUrl>	Primary + mirror URLs; first isPrimary=true is canonical. MirrorUrl is a lava.sdk.api primitive from the Tracker-SDK submodule.
capabilities	Set<TrackerCapability>	What the tracker actually supports (clause 6.E —

Field	Type	Meaning
		declarations of fact, not intent).
authType	AuthType	Auth mechanism (AuthType.kt).
encoding	String	e.g. "UTF-8", "Windows-1251".
expectedHealthMarker	String	Substring that must appear on the root page for a HEALTHY probe.
verified	Boolean (default false)	Clause 6.G gate — true only with a passing Challenge Test. Fail-closed: hidden from the provider list until earned.
supportsAnonymous	Boolean (default false)	Per-provider anonymous-browse capability (Phase 1.5, 2026-05-04).
apiSupported	Boolean (default false)	Whether lava-api-go routes /v1/{trackerId}/... for this provider today (Phase 1 α -hotfix, 2026-05-06).

The three boolean flags (verified, supportsAnonymous, apiSupported) all default to false — a **fail-closed** posture so a half-wired provider never reaches users until each gate is explicitly satisfied.

2.2 TrackerCapability — the single source of truth

```
graph LR
    subgraph "TrackerCapability enum (13 values)"
        SEARCH; BROWSE; FORUM; TOPIC; COMMENTS; FAVORITES
        TORRENT_DOWNLOAD; MAGNET_LINK; AUTH_REQUIRED
        CAPTCHA_LOGIN; RSS; UPLOAD; USER_PROFILE
    end
```

Verbatim from TrackerCapability.kt: SEARCH, BROWSE, FORUM, TOPIC, COMMENTS, FAVORITES, TORRENT_DOWNLOAD, MAGNET_LINK, AUTH_REQUIRED, CAPTCHA_LOGIN, RSS, UPLOAD, USER_PROFILE.

The enum is the **single source of truth** that TrackerDescriptor.capabilities and TrackerClient.getFeature<T>() are gated on. Adding a feature interface without an enum entry makes the interface unreachable — a 6.E violation (see core/CLAUDE.md "Scoped clause for new feature interfaces").

Note on RSS, UPLOAD, USER_PROFILE: these enum values exist but have **no dedicated TrackerFeature interface** in `core/tracker/api/feature/` today — they are reserved capability slots, **NOT IMPLEMENTED** as feature interfaces. The seven implemented feature interfaces are listed in §2.3.

2.3 The seven feature interfaces

Each interface extends the TrackerFeature marker (`core/tracker/api/.../TrackerFeature.kt`). Method signatures are verbatim from `core/tracker/api/src/main/kotlin/lava/tracker/api/feature/`:

```
classDiagram
    class TrackerFeature {
        <<marker>>
    }
    class SearchableTracker {
        +search(request, page) SearchResult
    }
    class BrowseableTracker {
        +browse(category, page) BrowseResult
        +getForumTree() ForumTree?
    }
    class TopicTracker {
        +getTopic(id) TopicDetail
        +getTopicPage(id, page) TopicPage
    }
    class CommentsTracker {
        +getComments(topicId, page) CommentsPage
        +addComment(topicId, message) Boolean
    }
    class FavoritesTracker {
        +list() List~TorrentItem~
        +add(id) Boolean
        +remove(id) Boolean
    }
    class AuthenticatableTracker {
        +login(req) LoginResult
        +logout()
        +checkAuth() AuthState
    }
    class DownloadableTracker {
        +downloadTorrentFile(id) ByteArray
        +getMagnetLink(id) String?
    }
    TrackerFeature <|-- SearchableTracker
    TrackerFeature <|-- BrowseableTracker
    TrackerFeature <|-- TopicTracker
    TrackerFeature <|-- CommentsTracker
    TrackerFeature <|-- FavoritesTracker
    TrackerFeature <|-- AuthenticatableTracker
    TrackerFeature <|-- DownloadableTracker
```

Interface	File	Capability it answers
SearchableTracker	feature/ SearchableTracker.kt	SEARCH
BrowsableTracker	feature/ BrowsableTracker.kt	BROWSE / FORUM
TopicTracker	feature/TopicTracker.kt	TOPIC
CommentsTracker	feature/ CommentsTracker.kt	COMMENTS
FavoritesTracker	feature/ FavoritesTracker.kt	FAVORITES
AuthenticatableTracker	feature/ AuthenticatableTracker.kt	AUTH_REQUIRED / CAPTCHA_LOGIN
DownloadableTracker	feature/ DownloadableTracker.kt	TORRENT_DOWNLOAD / MAGNET_LINK

`DownloadableTracker.getMagnetLink(id)` is the **only non-suspend** method on any feature interface, and its contract ("Returns null if the magnet URI is not synchronously available without an HTTP fetch") is the reason the per-provider **magnet-cache pattern** exists — see §6 and the developer guide.

3. Capability gate — `getFeature<T>()`

`TrackerClient` (`core/tracker/api/.../TrackerClient.kt`) is the runtime enforcement point for clause 6.E:

```
interface TrackerClient : AutoCloseable {
    val descriptor: TrackerDescriptor
    suspend fun healthCheck(): Boolean
    fun <T : TrackerFeature> getFeature(featureClass: KClass<T>):
        T?
}
```

The contract, stated in the interface's KDoc: *"capability declared in descriptor ⇒ this returns non-null."* The inverse is the load-bearing honesty guarantee — a capability **not** declared returns null, and every SDK call site treats null as a typed failure outcome rather than throwing.

sequenceDiagram

```
participant SDK as LavaTrackerSdk
participant C as TrackerClient (active)
participant D as TrackerDescriptor
participant F as SearchableTracker impl
```

```
SDK->>C: getFeature(SearchableTracker::class)
C->>D: SEARCH in descriptor.capabilities?
alt declared
    D-->>C: true
    C-->>SDK: SearchableTracker impl
```

```

        SDK-->>F: search(request, page)
        F-->>SDK: SearchResult
    else not declared
        D-->>C: false
        C-->>SDK: null
    Note over SDK: SearchOutcome.Failure<br/>("tracker X does
not support SEARCH")<br/>- never throws
end

```

This pattern repeats for every capability across LavaTrackerSdk (search, browse, getTopic, getCommentsPage, addComment, getFavorites, addFavorite, removeFavorite, login, checkAuth, logout, getMagnetLink, downloadTorrent) — each does a `getFeature(...)` null-check and returns a Failure/null/false outcome instead of throwing (LavaTrackerSdk.kt).

4. Registry + factory — how a provider gets wired

4.1 Registration topology (Hilt, core:tracker:client)

There is **one** Hilt module that builds the registry: `core/tracker/client/src/main/kotlin/lava/tracker/client/di/TrackerClientModule.kt`. Its `provideTrackerRegistry @Provides` function takes every provider's `TrackerClientFactory` as a constructor parameter and `register(...)`s each into a `DefaultTrackerRegistry`:

```

graph TD
    subgraph "TrackerClientModule.provideTrackerRegistry (Hilt @Provides)"
        RR["RuTrackerClientFactory"]
        RT["RuTorClientFactory"]
        NN["NmclubClientFactory"]
        KZ["KinozalClientFactory"]
        AO["ArchiveOrgClientFactory"]
        GB["GutenbergClientFactory"]
    end
    REG["DefaultTrackerRegistry<br/>(register each factory)"]
    RR --> REG
    RT --> REG
    NN --> REG
    KZ --> REG
    AO --> REG
    GB --> REG
    REG -->|"exposes by descriptor.id"| SDK["LavaTrackerSdk"]

```

The six factories registered today (verbatim from `TrackerClientModule.kt provideTrackerRegistry`): `rutrackerFactory`, `rutorFactory`, `nmclubFactory`, `kinozalFactory`, `archiveOrgFactory`, `gutenbergFactory`.

Doc-drift note: `docs/sdk-developer-guide.md §2 Step 5` still describes a `@IntoSet`-multibinding registration style and §5 lists only `RuTracker` + `RuTor`. The **actual** wiring in `TrackerClientModule.kt` is explicit

constructor injection of six factories into provideTrackerRegistry (no @IntoSet). This diagram reflects the code. The guide's Step 5 is documentation that predates the additional providers.

4.2 TrackerClientFactory — clone-aware creation

TrackerClientFactory (core/tracker/registry/.../TrackerClientFactory.kt) extends the SDK primitive PluginFactory<TrackerDescriptor, TrackerClient> and exposes create(config: PluginConfig): TrackerClient.

The config carries an optional clone base-URL override (PluginConfig.cloneBaseUrlOverride, read via lava.tracker.registry.cloneBaseUrlOverride). When present (SP-4 Phase F.2), create returns a **per-call** client routed at the clone's primaryUrl; otherwise it returns the shared @Singleton client. See KinozalClientFactory.kt for the canonical implementation — note it shares ONE KinozalMagnetCache between the clone's topic + download features (§6).

5. Request flow: app → SDK → client → feature

End-to-end search flow through the production classes:

sequenceDiagram

```
actor User
participant VM as feature:search_result ViewModel
participant SDK as LavaTrackerSdk
participant REG as DefaultTrackerRegistry
participant FAC as KinozalClientFactory
participant C as KinozalClient
participant S as KinozalSearch (SearchableTracker)
participant MC as KinozalMagnetCache
```

```
User->>VM: type query, tap Search
VM->>SDK: multiSearch(request, providerIds)
SDK->>REG: get("kinozal", config)
REG->>FAC: create(config)
FAC-->>SDK: KinozalClient
SDK->>C: getFeature(SearchableTracker::class)
C-->>SDK: KinozalSearch (capability declared)
SDK->>S: search(request, page)
S->>S: HTTP GET + parse rows
S->>MC: put(torrentId, magnetUri) per row
S-->>SDK: SearchResult
SDK->>SDK: DeduplicationEngine.deduplicate(...)
SDK-->>VM: UnifiedSearchResult
VM-->>User: render result rows
```

Grounding for the multi-provider path: LavaTrackerSdk.multiSearch / streamMultiSearch fan out across providerIds with parallel async { ... }.awaitAll() (SP-4 Phase D), roll up per-provider statuses, and deduplicate

via DeduplicationEngine (core/tracker/client/.../dedup/
DeduplicationEngine.kt). The magnet-cache write on each search row
(S->>MC: put) is the real code in NnmclubSearch.kt (result.items.forEach
{ magnetCache.put(it.torrentId, it.magnetUri) }).

Cross-tracker / mirror fallback

The single-active-tracker path (search, browse) additionally runs through
runWithMirrorFallback, which (when mirror collaborators are wired) executes
via a per-tracker MirrorManager and, on total mirror exhaustion, proposes a
CrossTrackerFallbackProposed outcome via CrossTrackerFallbackPolicy
(core/tracker/client/.../CrossTrackerFallbackPolicy.kt). There is **no
silent fallback** — the UI shows a modal and the user accepts or dismisses (see
docs/ARCHITECTURE.md "Cross-tracker fallback flow").

The switchTracker / activeTrackerId single-active-tracker API is
@Deprecated (SP-4 Phase D) in favor of multiSearch /
streamMultiSearch with explicit provider lists (LavaTrackerSdk.kt).

6. The magnet-cache pattern (§6.E Capability Honesty)

DownloadableTracker.getMagnetLink(id) is **non-suspend**, so it cannot make
an HTTP call. Its contract is to return the magnet **only if synchronously available**,
else null. Three providers (kinozal, nmclub, rutor) declare the MAGNET_LINK
capability and back getMagnetLink with a @Singleton magnet cache that the
suspend topic/search path populates from genuinely-parsed magnets:

```
graph LR
    subgraph "write side (suspend, real fetch)"
        TOPIC["KinozalTopic.getTopic(id)<br/>parses topic page"]
        SEARCH["NnmclubSearch.search(...)<br/>parses result rows"]
    end
    end
    CACHE["@Singleton XxxMagnetCache<br/>ConcurrentHashMap<id, magnet>"]
    subgraph "read side (non-suspend)"
        DL["KinozalDownload.getMagnetLink(id)<br/>magnetCache.get(id)"]
    end
    end
    TOPIC -->|"put(id, magnetUri)"| CACHE
    SEARCH -->|"put(torrentId, magnetUri)"| CACHE
    CACHE -->|"get(id) → real magnet OR null"| DL
```

Real files implementing this pattern:

Provider	Cache	Writer(s)	Reader
Kinozal	kinozal/.../magnet/ KinozalMagnetCache.kt	KinozalTopic.getTopic	KinozalDownload.getMagnetLink
NNM-Club	nmclub/.../http/ NnmclubMagnetCache.kt	NnmclubTopic, NnmclubSearch	NnmclubDownload.getMagnetLink

Provider	Cache	Writer(s)	Reader
RuTor	rutor/.../magnet/ RuTorMagnetCache.kt	RuTor topic/search parsers	RuTor download feature

Why this is honest rather than a bluff: `get(id)` returns `null` when an `id` was never surfaced (an honest "not synchronously available" per the contract), and the cache is only ever populated with magnets that the **production parser** extracted from a real page — never fabricated. The cache is a `ConcurrentHashMap` (search + topic features may write from different coroutines while download reads). It is `@Singleton` so all features share one instance; the clone path passes an explicit shared instance (`KinozalClientFactory.create`, §4.2).

The full author-facing recipe — including the put-on-parse wiring and the clone-path shared-instance rule — is documented in [docs/sdk-developer-guide.md §8](#).

7. Proving a download is real — the torrent validators

`core:common` ships two validators (`core/common/src/main/kotlin/lava/common/torrent/`) that turn "we returned bytes / a string" into "we returned a genuinely-usable download". Both produce a `DownloadValidationResult` { `valid`, `infoHashHex`, `reason` } keyed on the **same** lowercase 40-char info-hash, so a `.torrent` file and a magnet for the same content cross-check.

```
graph TD
    BYTES[".torrent bytes (ByteArray)"]
    MAGNET["magnet: URI (String)"]
    TFV["TorrentFileValidator.validate(bytes)<br/>bencode → info dict → SHA-1(info)"]
    MLV["MagnetLinkValidator.validate(uri)<br/>xt=urn:btih:<hex|base32> → 40-hex"]
    RES["DownloadValidationResult<br/>{ valid, infoHashHex, reason }"]
    BYTES --> TFV
    MAGNET --> MLV
    TFV --> RES
    MLV --> RES
    RES -->|"same content ⇒ same infoHashHex"| RES
```

- **TorrentFileValidator** — parses the bytes as bencode (`Bencode.kt`), requires a top-level dict containing `info` with a non-empty name, a piece length `> 0`, and pieces whose length is a positive multiple of 20 (each piece SHA-1 is 20 bytes). The info-hash is SHA-1 over the **verbatim** bencoded info dict bytes (recovered via `valueRanges`), so it matches what a real BitTorrent client computes.
- **MagnetLinkValidator** — requires an `xt=urn:btih:<hash>` where `<hash>` is 40-char hex OR 32-char RFC-4648 base32; normalizes both to lowercase hex.

These are the canonical anti-bluff way to assert a download feature actually works (clause 6.E + the Anti-Bluff Pact's "primary assertion on user-visible state"). See [docs/sdk-developer-guide.md §9](#).

8. Module layout

```
graph TD
    API["core:tracker:api<br/>feature interfaces + capability enum  
+ DTOs"]
    REG["core:tracker:registry<br/>TrackerRegistry over  
lava.sdk:registry"]
    CLIENT["core:tracker:client<br/>LavaTrackerSdk + Hilt module +  
persistence"]
    COMMON["core:common<br/>torrent validators + bencode"]
    subgraph "per-provider plugins (core:tracker:*)"
        RUTRACKER["rutracker"]
        RUTOR["rutor"]
        KINOZAL["kinozal"]
        NNMCCLUB["nnmclub"]
        ARCHIVEORG["archiveorg"]
        GUTENBERG["guttenberg"]
    end
    SUB["submodules/tracker_sdk<br/>(vasic-digital/Tracker-  
SDK)<br/>MirrorUrl, PluginRegistry, PluginFactory"]

    RUTRACKER --> API
    RUTOR --> API
    KINOZAL --> API
    NNMCCLUB --> API
    ARCHIVEORG --> API
    GUTENBERG --> API
    KINOZAL --> COMMON
    NNMCCLUB --> COMMON
    RUTOR --> COMMON
    REG --> API
    REG --> SUB
    CLIENT --> REG
    CLIENT --> RUTRACKER
    CLIENT --> RUTOR
    CLIENT --> KINOZAL
    CLIENT --> NNMCCLUB
    CLIENT --> ARCHIVEORG
    CLIENT --> GUTENBERG
    API --> SUB
```

Generic primitives (MirrorUrl, PluginRegistry, PluginFactory, MapPluginConfig) live in the vasic-digital/Tracker-SDK submodule mounted at submodules/tracker_sdk/ per the Decoupled Reusable Architecture rule; the pin is frozen by default. DefaultTrackerRegistry composes the SDK's DefaultPluginRegistry via Kotlin interface delegation (the SDK class is final) and adds trackersWithCapability(...).

9. Where to go next

- **Add a provider:** [docs/sdk-developer-guide.md](#) (7-step recipe + §8 magnet cache + §9 validators).
- **Server-side deployment / Jackett:** [docs/architecture/local-stack-topology.md](#).
- **mDNS discovery:** [docs/LOCAL_NETWORK_DISCOVERY.md](#).

Last updated: 2026-06-08. All symbols verified against the working tree.